Recommendations — aml-mds-related-rr-tp53-aberrant-hla-pending-x7q2-rerun2

Ranked therapeutic options with likelihood, toxicity, mechanism risk, and key references

Generated 2026-08-07

Recommendations — aml-mds-related-rr-tp53-aberrant-hla-pendi

Profile snapshot

- **Primary site:** bone marrow (hematologic)
- **Histology:** Acute myeloid leukemia, myelodysplasia-related (AML-MR), relapsed/refractory; transformed from prior MDS (RCMD, IPSS-R high/very-high, diagnosed ~2019); with extramedullary myeloid sarcoma
- **Stage:** Relapsed/refractory AML-MR. Marrow ~85% blasts (June 2026); peripheral-blood blasts 58->82% (late June 2026); extramedullary right-thigh myeloid sarcoma on MRI (May 2026, adverse feature). Refractory to FLAG-IDA + venetoclax reinduction. Late relapse ~6 years after matched-sibling PBSC allo-HCT (~2020); pattern consistent with graft-versus-leukemia escape rather than graft failure.
- **ECOG:** 1
- **Age band:** 50-59
- **Sex:** female
- **Biomarkers:** TP53 aberrant, pending confirmation (NOT resolved in either direction). Pathogenic TP53 mutation reported on 2019 diagnostic NGS; 2026 relapse NGS reported NEGATIVE, but on a limited assay (LOD ~2% VAF, no copy-number/CNV detection, not MRD-grade). Discrepancy most plausibly subclonal loss, copy-loss, or a re-review artifact rather than true reversion. Working call: TP53-aberrant pending confirmation. (ngs_pending); TP53 Y220C (specific variant) unknown / to be determined — Y220C status not established; therapy handle (rezatapopt) is contingent on it (ngs_pending); HLA-A*02:01 unknown / pending retrieval. High-resolution HLA typing was performed ~2020 but the allele-level result is not in the records on hand (placeholder in the EHR). (unknown); HLA-A*24:02 unknown / pending (unknown); HA-1 / HA-2 minor histocompatibility antigens (recipient and sibling donor) unknown / pending for BOTH patient and sibling donor. HLA-identical siblings can still differ at HA-1/HA-2. (unknown); CD33 positive (blasts CD33+ by flow cytometry); CD123 positive (blasts CD123+ by flow cytometry); AML immunophenotype (supporting) CD34+ (increased), CD13+, CD15+ (subset), CD117+ (variably increased), HLA-DR (variably decreased), CD38 decreased; negative CD4/CD5/CD7/CD14/CD16/CD19/CD56/CD64; FLT3 negative / no mutation (2019 diagnostic NGS); IDH1 negative / no mutation (2019 diagnostic NGS); IDH2 negative / no mutation (2019 diagnostic NGS); JAK3 variant of uncertain significance (VUS) — 2019 (unknown); SMC3 variant of uncertain significance (VUS) — 2019 (cohesin-complex gene) (unknown); Cytogenetics / karyotype Complex, adverse-risk. 2019: complex karyotype with del(5q), del(7q), del(20q12), der(17)t(13;17) [17p], monosomy 13, others. 2026 relapse: gain of 5' MECOM, del(5q), del(7q), KMT2A amplification (4 copies), del(8cen), del(16q22). Shared del(5q)/del(7q) + TP53 founder across timepoints indicates evolution of the ORIGINAL clone transformed to AML (not a new/secondary or donor-derived leukemia). (ngs_pending)
- **Efficacy/toxicity weight:** 0.75

27 ranked therapeutic options across 7 targetable features. Biomarker workup steps live in the standalone Target validation paths report.

Cd33 interventions

1. CLAG-GO: cladribine, high-dose cytarabine and G-CSF with fractionated gemtuzumab ozogamicin 3 mg/m2 on days 1, 4 and 7 (NCT04050280)

Likelihood of effect: Moderate as cytoreduction: CLAG backbone with a CD33 payload, but no published CLAG-GO efficacy in refractory AML and Hills 2014 null in her karyotype.

Toxicity burden: High (prolonged grade 4 neutropenia and thrombocytopenia, febrile neutropenia and infection, calicheamicin hepatotoxicity including veno-occlusive disease)

Counter-productive MoA:Moderate (Nucleoside cross-resistance after FLAG-IDA failure; prolonged aplasia in a six-year-old graft could end the transplant plan rather than enable it.)

Overall:The gate on almost everything else here: the one strong-fit open trial accepting post-allograft relapse, with a CD33 component the randomised data do not support in her karyotype.

Key references: NCT04050280 · PMID 17051246 · PMID 25008258

2. BMS-986497 (ORM-6151), CD33-directed antibody-conjugated GSPT1 degrader, monotherapy arm (NCT06419634)

Likelihood of effect: Unknown: first-in-human with no published clinical data. Listed because it is open, reachable and independent of the pending HLA answer.

Toxicity burden: Moderate (no published clinical safety; CD33-ADC and GSPT1-degrader class myelosuppression and hepatic risk expected)

Counter-productive MoA:Low (On-target CD33 killing of normal myeloid progenitors could deepen aplasia and delay the graft; no mechanism objection was raised.)

Overall:The CD33 trial fallback that does not depend on the pending allele, with unusually open posted criteria and no clinical data behind it whatsoever.

Key references: NCT06419634

Also considered — not ranked

Feature-targeting investigational options the board considered but did not rank as live top-tier choices: Unavailable (program discontinued), Consolidated (one product of an approach ranked above), Thin evidence (no peer-reviewed clinical efficacy yet), or Not enrollable (cross-tumor or geographically inaccessible).

CD33-directed CAR-T (CART-33 and successors, NCT05672147) — _Thin evidence_

Likelihood of effect: Thin: one published patient with a two-week blast reduction and progression by nine weeks. Recruiting protocol permits her prior transplant.

Toxicity burden: Moderate (single published patient with fever and cytokine elevation; CAR-T class cytokine release and lymphodepletion cytopenias expected)

Counter-productive MoA:Moderate (On-target CD33 killing of normal myeloid progenitors could deepen aplasia in a marrow that must support a graft.)

Overall:A recruiting CD33 CAR-T that accepts her prior transplant, resting on a single published patient who progressed at nine weeks.

Key references: PMID 25174587 · NCT05672147

GTB-3650 (anti-CD16 / IL-15 / anti-CD33 tri-specific killer engager, NCT06594445) — _Not enrollable_

Likelihood of effect: Not enrollable: the protocol excludes transplant candidates, which is the entire plan here. No published clinical efficacy.

Toxicity burden: Low (no published clinical toxicity)

Counter-productive MoA:Low (NK-cell engagement spares the T-cell compartment; no mechanism objection raised.)

Overall:A CD33 NK-engager that excludes transplant candidates by protocol, which rules out a case built around a second graft.

Key references: NCT06594445

Dual CD33/CD123-directed CAR-T (CLL-1/CD33/CD123 constructs, NCT04010877) — _Not enrollable_

Likelihood of effect: Not practically enrollable: single non-US site with no published clinical efficacy, despite hitting both stated antigen handles at once.

Toxicity burden: Moderate (no published grade profile; CAR-T class cytokine release and dual-antigen on-target myelosuppression expected)

Counter-productive MoA:Low (Dual-antigen design limits single-antigen escape; on-target myelosuppression across two myeloid antigens is the trade.)

Overall:The dual CD33/CD123 construct the escape-hedging logic calls for, at a single non-US site with no published clinical data.

Key references: NCT04010877

Lintuzumab-Ac225 (225Ac anti-CD33, Actimab-A) after CLAG-M salvage — _Unavailable_

Likelihood of effect: Not obtainable (supply suspension): composite CR 56.6% on n=26 belongs to CLAG-M plus the alpha-emitter, not to the rank-2 regimen.

Toxicity burden: High (grade 3-4 febrile neutropenia 65.4%, white blood cell decrease 50%, prolonged cytopenias)

Counter-productive MoA:Low (Alpha-emitter marrow injury ahead of a planned graft; no mechanism objection was raised by the board.)

Overall:The source of the 56.6% figure that keeps being attached to the rank-2 regimen, on a drug whose trials are suspended for supply.

Key references: PMID 39955432 · NCT02575963

Evidence in detail

CLAG-GO: cladribine, high-dose cytarabine and G-CSF with fractionated gemtuzumab ozogamicin 3 mg/m2 on days 1, 4 and 7 (NCT04050280)

Taksin et al. *Leukemia* 2007

- **Disease context:AML in first relapse** (2L+) — 57 patients in first relapse. Remission correlated with P-glycoprotein and MRP1 activity, which is the drug-efflux axis rather than CD33 density.
- **n:** 57
- **Toxicities:Neutropenia <500/microL** (grade >=3): (n=57); **Thrombocytopenia <50,000/microL** (grade >=3): (n=57); **Hepatic toxicity** (grade >=3): 0.0% (n=57); **Veno-occlusive disease** (grade any): 0.0% (n=57)
- **Efficacy:CR_rate** 26%; **ORR** 33%; **RFS** 11 months

Hills et al. *Lancet Oncology* 2014

- **Disease context:Adult AML (excluding acute promyelocytic leukaemia)** (1L) — 3325 patients across five randomised trials, with a prespecified exploratory stratification by cytogenetic risk group.
- **n:** 3325
- **Toxicities:Early death** (grade 5)
- **Efficacy:OS_rate** 0.9 OR (95% CI 0.82-0.98); **other** 0.81 OR (95% CI 0.73-0.9); **CR_rate** 0.91 OR (95% CI 0.77-1.07); **OS_rate** 0.47 OR (95% CI 0.31-0.73); **OS_rate** 0.84 OR (95% CI 0.75-0.95); **OS_rate** 0.99 OR (95% CI 0.83-1.18)

CD33-directed CAR-T (CART-33 and successors, NCT05672147)

Wang et al. *Molecular Therapy* 2015

- **Disease context:Refractory AML** (2L+) — Single 41-year-old male patient with refractory AML and pre-existing pancytopenia.
- **n:** 1
- **Toxicities:Chills and fever** (grade any): 100.0% (n=1); **Cytokine elevation (IL-6, IL-8, TNF-alpha, IFN-gamma)** (grade any): 100.0% (n=1); **Hyperbilirubinaemia** (grade 1-2): 100.0% (n=1); **Pancytopenia fluctuation** (grade >=3): 100.0% (n=1)
- **Efficacy:other** marked marrow blast decrease at 2 weeks—; **TTP** 2.1 months

Lintuzumab-Ac225 (225Ac anti-CD33, Actimab-A) after CLAG-M salvage

Abedin et al. *Leukemia* 2025

- **Disease context:Relapsed/refractory AML** (2L+) — 26 patients (21 in 3+3 escalation, 5 at the RP2D); 86.7% of evaluable patients had high-risk disease. Separate readouts for TP53-mutated and prior-venetoclax subgroups.
- **n:** 26
- **Toxicities:Febrile neutropenia** (grade 3-4): 65.4% (n=26); **White blood cell count decreased** (grade 3-4): 50.0% (n=26)
- **Efficacy:CR_rate** 56.6%; **CR_rate** 50%; **CR_rate** 38.5%; **OS_rate** 23.1% (95% CI 9.4-40.3); **PFS_rate** 30.8% (95% CI 14.6-48.5); **biomarker** 66.7%

Hct2 Platform interventions

1. Second allogeneic HCT (HCT2) from the HLA-identical sibling on a cytoreduce-first or sequential-conditioning platform (e.g. DEC-FLAG-Ida/RIC, NCT06928662)

Likelihood of effect: Moderate if cytoreduced, low as she stands: four registries replicate 2-year OS 37% +/- 6% in her donor stratum, all gated on blasts under 27-35%.

Toxicity burden: High (non-relapse mortality after a second graft, acute and chronic GVHD, prolonged cytopenias, infection; no registry row tabulates per-organ adverse events)

Counter-productive MoA:Moderate (The same sibling repertoire that already lost control; HLA loss or class II downregulation would defeat graft-versus-leukaemia a second time.)

Overall:The only documented cure fraction here, backed by four registries replicating a prognostic model rather than a treatment effect, and gated entirely on getting blasts under 27%.

Key references: PMID 26367221 · PMID 23918951 · PMID 22167752

2. Sibling donor lymphocyte infusion after cytoreduction (EBMT therapeutic starting dose 1x10^7 CD3/kg; interferon-gamma-1b + DLI on NCT06529731 as the trial route)

Likelihood of effect: Low as she stands, moderate after cytoreduction: Schmid 2007 puts active-disease DLI at 15% +/- 3% two-year survival against 56% +/- 10% in remission.

Toxicity burden: Moderate (acute and chronic GVHD, marrow aplasia, infection; no registry row grades adverse events, so the harm side is unquantified)

Counter-productive MoA:Moderate (Re-infusing the donor repertoire that already stopped holding this disease; grade III-IV GVHD would close the door on HCT2.)

Overall:The lower-intensity route to graft-versus-leukaemia if fitness rules out a second conditioning regimen, but off-guideline at 85% blasts and shut out of its own trial until the marrow comes down.

Key references: PMID 17909197 · PMID 29526774 · NCT06529731

3. 211At-BC8-B10 (astatine-211 anti-CD45) targeted conditioning ahead of haploidentical allogeneic HCT (NCT03670966)

Likelihood of effect: Unknown for this construct: no human efficacy published. The borrowed SIERRA signal (dCR 17.1% vs 0%) came from a population that excluded prior allo-HCT.

Toxicity burden: High (alpha-emitter conditioning plus a second graft: prolonged aplasia, infection and non-relapse mortality; no human adverse-event table published for this construct)

Counter-productive MoA:Moderate (Alpha-emitter marrow injury before a second graft in a heavily pretreated marrow; needs a haploidentical relative, not the existing sibling.)

Overall:The one targeted-conditioning protocol that admits her prior graft, carrying a principle SIERRA proved in a population that excluded her and no human efficacy of its own.

Key references: NCT03670966 · PMID 39298738 · PMID 23471305

Also considered — not ranked

Feature-targeting investigational options the board considered but did not rank as live top-tier choices: Unavailable (program discontinued), Consolidated (one product of an approach ranked above), Thin evidence (no peer-reviewed clinical efficacy yet), or Not enrollable (cross-tumor or geographically inaccessible).

131I-apamistamab (Iomab-B) anti-CD45 radioimmunotherapy conditioning before allogeneic HCT — _Unavailable_

Likelihood of effect: Not obtainable: the only phase 3 to hit its primary endpoint (dCR 17.1% vs 0%) excluded prior allo-HCT, and the successor excludes it again.

Toxicity burden: High (grade 3+ treatment-related adverse events 59.7%, at parity with conventional care; conditioning cytopenias, infection, mucositis)

Counter-productive MoA:Low (Radioconjugate marrow injury before engraftment; the trial-level problem is flat overall survival, not mechanism antagonism.)

Overall:The strongest design tier in the dossier, closed to her on both protocols and with flat overall survival even where the durable-CR difference was stark.

Key references: PMID 39298738 · NCT02665065 · NCT07157514

FLAMSA-RIC sequential conditioning (fludarabine/cytarabine/amsacrine into reduced-intensity conditioning; US analogues DEC-FLAG-Ida/RIC on NCT06928662 and FLAMSAClax on NCT05807932) — _Consolidated_

Likelihood of effect: Consolidated into rank 3: the sequential-conditioning form of the ranked transplant approach. CR 88% (66/75) in largely refractory disease, 42% alive at two years.

Toxicity burden: High (second-graft non-relapse mortality, GVHD, prolonged cytopenias, infection)

Counter-productive MoA:Moderate (Repeating the FLAG-Ida backbone she progressed through risks buying the conditioning toxicity without the cytoreduction.)

Overall:The sequential-conditioning route into the ranked second transplant: reachable in the US only as DEC-FLAG-Ida/RIC or an institutional regimen, and it repeats a backbone she failed.

Key references: PMID 16110027 · NCT06928662 · NCT05807932

Evidence in detail

Second allogeneic HCT (HCT2) from the HLA-identical sibling on a cytoreduce-first or sequential-conditioning platform (e.g. DEC-FLAG-Ida/RIC, NCT06928662)

Ruutu et al. *Bone Marrow Transplantation* 2015

- **Disease context:Relapse of malignant disease after a first allogeneic transplant** (2L+) — 2632 second

allogeneic transplants performed for relapse after a first transplant, across diagnoses. Sibling donor was a favourable factor; changing donor was not.
- **n:** 2632
- **Toxicities:** Adverse events are not tabulated in this registry analysis; the outcome measures are survival and relapse.
- **Efficacy:RFS** 15%; **other** favourable—

Christopeit et al. *Journal of Clinical Oncology* 2013

- **Disease context:Acute leukemia relapsing after a first allogeneic HCT** (2L+) — 179 second transplants, 75 after a matched related first transplant and 104 after an unrelated first transplant. Remission duration after the first transplant and disease stage at the second were the two prognostic factors carried into multivariate analysis.
- **n:** 179
- **Toxicities:** The report is survival-focused and does not tabulate per-term adverse events; failure was driven by second relapse in about half of those achieving CR.
- **Efficacy:OS_rate** 25%; **OS_rate** 37%; **OS_rate** 16%; **HR_OS** 2.37 HR (95% CI 1.61-3.46); **HR_OS** 0.53 HR (95% CI 0.34-0.83); **HR_OS** 0.68 HR (95% CI 0.47-0.98); **CR_rate** 74%

Schmid et al. *Blood* 2012

- **Disease context:AML relapsing after reduced-intensity conditioning allogeneic HCT** (2L+) — 263 relapsed patients drawn from 2815 RIC transplants performed in CR between 1999 and 2008. Prognostic groups were built from remission duration after HCT, marrow blast percentage and prior acute GVHD.
- **n:** 263
- **Toxicities:** Survival, response and prognostic grouping are the reported outcomes; no per-term adverse-event data appear in the paper.
- **Efficacy:OS_rate** 14%; **CR_rate** 32%; **HR_OS** 0.5 HR (95% CI 0.37-0.67); **HR_OS** 0.53 HR (95% CI 0.4-0.72); **HR_OS** 0.67 HR (95% CI 0.49-0.93); **OS_rate** 32%; **OS_rate** 4%

Sibling donor lymphocyte infusion after cytoreduction (EBMT therapeutic starting dose 1x10^7 CD3/kg; interferon-gamma-1b + DLI on NCT06529731 as the trial route)

Schmid et al. *Journal of Clinical Oncology* 2007

- **Disease context:AML in first haematologic relapse after allogeneic HCT** (2L+) — 399 patients in first relapse after HCT; 171 received DLI and 228 did not. Risk factors examined within the DLI group included marrow blast burden, karyotype, sex and remission status at infusion.
- **n:** 399
- **Toxicities:** The analysis reports survival and risk factors rather than a per-term adverse-event table; GVHD after DLI is discussed narratively as the mechanism of both benefit and harm.
- **Efficacy:OS_rate** 21%; **OS_rate** 9%; **OS_rate** 56%; **OS_rate** 15%; **other** favourable—; **other** favourable—

Ortí et al. *Experimental Hematology* 2018

- **Disease context:Acute leukemia relapsing after allogeneic HCT** (2L+) — 46 patients (30 AML, 16 ALL), median age 38; 27 received a second allograft and 19 received DLI, all after debulking.
- **n:** 46
- **Toxicities:** Comparative survival and relapse are the reported outcomes; the paper does not tabulate adverse events by term.
- **Efficacy:OS** higher with second allo-HCT on univariate analysis—; **DFS** trend favouring second allo-HCT—;

other time to relapse was the only significant predictor of OS and DFS—

211At-BC8-B10 (astatine-211 anti-CD45) targeted conditioning ahead of haploidentical allogeneic HCT (NCT03670966)

Gyurkocza et al. *Journal of Clinical Oncology* 2025

- **Disease context:Active relapsed/refractory AML, age 55 and over** (2L+) — 153 patients randomised (76 apamistamab, 77 conventional care). Active disease at randomisation was required. Prior allogeneic or autologous HSCT was an exclusion, so this patient could not have enrolled.
- **n:** 153
- **Toxicities:Any treatment-related AE** (grade >=3): 59.7% (n=76); **Febrile neutropenia** (grade >=3): 18.1% (n=76); **Mucositis** (grade >=3): 15.2% (n=66); **Acute GVHD** (grade 1-2): 27.3% (n=66); **Acute GVHD** (grade >=3): 9.1% (n=66); **Sepsis** (grade >=3): 6.1% (n=66); **Non-relapse mortality (100 day)** (grade 5): 12.2% (n=76); **Treatment-related death** (grade 5): 4.2% (n=72)
- **Efficacy:CR_rate** 17.1% (95% CI 9.4-27.5); **CR_rate** 0.0% (95% CI 0.0-4.7); **HR_OS** 0.99 HR (95% CI 0.7-1.41); **HR_EFS** 0.23 HR (95% CI 0.15-0.34); **TRAE_rate** 59.7%

Orozco et al. *Blood* 2013

- **Disease context:** —
- **n:** —
- **Toxicities:** —
- **Efficacy:** —

131I-apamistamab (Iomab-B) anti-CD45 radioimmunotherapy conditioning before allogeneic HCT

Gyurkocza et al. *Journal of Clinical Oncology* 2025

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FLAMSA-RIC sequential conditioning (fludarabine/cytarabine/amsacrine into reduced-intensity conditioning; US analogues DEC-FLAG-Ida/RIC on NCT06928662 and FLAMSAClax on NCT05807932)

Schmid et al. *Journal of Clinical Oncology* 2005

- **Disease context:High-risk AML and MDS, including progressive or refractory disease** (2L+) — 75 consecutive patients, median age 52.3. 59 had progressive or refractory disease; unfavourable karyotype in 49% of informative patients; 68 had contraindications to standard conditioning. 31 had HLA-identical sibling donors.
- **n:** 75

- **Toxicities:** The report gives no per-term adverse-event table; outcome was driven by non-relapse mortality and GVHD, with survival best in patients receiving high CD34+ cell doses and having limited GVHD.
- **Efficacy:CR_rate** 88%; **OS_rate** 42%; **EFS** 40%

Hla A0201 interventions

1. HA-1 TCR-T, donor-derived CD4+/CD8+ memory T cells after fludarabine/cyclophosphamide lymphodepletion (Bleakley/HighPassBio platform, NCT03326921), as post-cytoreduction or post-graft consolidation *(conditional on hla_a0201 positive)*

Likelihood of effect: Assuming A*02:01 and the HA-1 mismatch confirm: uncertain but curative-intent - 4/9 CR on a feasibility-and-DLT primary endpoint, and no data at 85% blasts.

Toxicity burden: Moderate (grade 3-5 infection after fludarabine/cyclophosphamide lymphodepletion, one grade 4 infusion reaction in 9; no CRS, no ICANS, no dose-limiting toxicity)

Counter-productive MoA:Moderate (Tumour burden blunts TCR-T; HLA loss or class I downregulation on relapsed blasts would leave the product blind to HA-1.)

Overall:The lever the referral was submitted to find, and the only HA-directed protocol requiring her prior graft rather than excluding it - live only if the typing confirms.

Key references: PMID 38683966 · NCT03326921 · PMID 29051183

2. CBX-250, CG1/HLA-A*02:01 x CD3 TCR-mimetic bispecific T-cell engager (CROSSCHECK-001, NCT06994676) *(conditional on hla_a0201 positive)*

Likelihood of effect: Unknown: first-in-human escalation with no efficacy readout. The argument is eligibility fit, eleven US sites and a compassionate-use route, not evidence.

Toxicity burden: Moderate (no human safety data; T-cell-engager class cytokine release and infusion reactions expected at high blast burden)

Counter-productive MoA:Moderate (Rides HLA-A*02:01 alone, so HLA loss silences it; CG1 expression on these blasts has never been measured.)

Overall:The A*02:01 option that needs nothing from the sibling and has a compassionate-use route, ranked on eligibility fit and reachability because there is no human efficacy data at all.

Key references: NCT06994676 · PMID 39939820 · PMID 40437170

Also considered — not ranked

Feature-targeting investigational options the board considered but did not rank as live top-tier choices: Unavailable (program discontinued), Consolidated (one product of an approach ranked above), Thin evidence (no peer-reviewed clinical efficacy yet), or Not enrollable (cross-tumor or geographically inaccessible).

HA-1 / HA-2 TCR-T commercial and next-generation products (TSC-100 and TSC-101 on ALLOHA/NCT05473910, BSB-1001 on NCT06704152, MDG1021 on the withdrawn NCT04464889) — _Not enrollable_

Likelihood of effect: Not enrollable: prior allo-HCT is an exclusion on every commercial HA-directed protocol; the HA-2 arm additionally rests on a different TCR.

Toxicity burden: Low (no clinical data; class safety carried by the rank-4 Bleakley platform)

Counter-productive MoA:Low (Antigen-loss escape possible; lineage-restricted killing limits collateral. No mechanism objection raised.)

Overall:The HA-directed products she cannot have: all three exclude prior allogeneic HCT, which is exactly what the rank-4 protocol requires.

Key references: NCT05473910 · NCT06704152 · PMID 42392183

WT1-directed HLA-A*02:01-restricted TCR-T (WT1-TCRC4 / TTCR-C4 on the terminated NCT01640301; successor FH-WT1-E50 + azacitidine on NCT07645469) — _Not enrollable_

Likelihood of effect: Not enrollable at 85% blasts: the open successor requires morphologic remission with MRD, and the active-disease cohort showed median OS 242 days.

Toxicity burden: Low (no dose-limiting toxicity and no product-related serious adverse events across n=15; grade 3 acute GVHD in the prophylaxis cohort)

Counter-productive MoA:Moderate (The class did little against active disease; WT1 is also broadly expressed on normal progenitors, risking on-target myelosuppression.)

Overall:The WT1 arm of the A*02:01 axis, and the cohort that supplies the tumour-burden argument used against ranking cellular therapy ahead of cytoreduction.

Key references: PMID 31235963 · PMID 40473616 · NCT07645469

MDG1011 (PRAME-directed, HLA-A*02:01-restricted autologous TCR-T) and BPX-701 (PRAME TCR-T with an iCasp9 switch) — _Unavailable_

Likelihood of effect: Not obtainable: both PRAME TCR-T programmes discontinued. MDG1011 reached 22.2% measurable response in 9 patients before stopping.

Toxicity burden: Low (no dose-limiting toxicity across n=9; cytokine release in 2 of 9)

Counter-productive MoA:Moderate (The sponsor's own reading was that tumour burden limited activity, the same failure mode that shapes the rank-4 sequencing.)

Overall:The PRAME half of the stated HLA-restricted handle, closed on both sides - and the source of the tumour-burden argument that set the sequencing.

Key references: PMID 41008812 · NCT03503968 · NCT02743611

Evidence in detail

HA-1 TCR-T, donor-derived CD4+/CD8+ memory T cells after fludarabine/cyclophosphamide lymphodepletion (Bleakley/HighPassBio platform, NCT03326921), as post-cytoreduction or post-graft consolidation

Krakow et al. *Blood* 2024

- **Disease context:Persistent or recurrent leukemia / myeloid neoplasm after allogeneic HCT** (2L+) — 9 allo-HCT recipients with disease recurrence after transplant, recipient HA-1(H)-positive and HLA-A*02:01-positive with an HA-1-disparate donor. Prior allo-HCT is an entry requirement rather than an exclusion.
- **n:** 9
- **Toxicities:CRS** (grade any): 0.0% (n=9); **ICANS** (grade any): 0.0% (n=9); **Dose-limiting toxicity** (grade >=3): 0.0% (n=9); **Acute GVHD (mild)** (grade 1-2): 22.2% (n=9); **Neutropenia** (grade >=3): (n=9); **Infection** (grade >=3): (n=9); **Infusion reaction** (grade 4): 11.1% (n=9)
- **Efficacy:CR_rate** 44.4%; **DoR** >27 months; **AE_rate** 0%

HA-1 / HA-2 TCR-T commercial and next-generation products (TSC-100 and TSC-101 on ALLOHA/NCT05473910, BSB-1001 on NCT06704152, MDG1021 on the withdrawn NCT04464889)

Krakow et al. *Blood Advances* 2026

- **Disease context:** —
- **n:** —
- **Toxicities:** —
- **Efficacy:** —

WT1-directed HLA-A*02:01-restricted TCR-T (WT1-TCRC4 / TTCR-C4 on the terminated NCT01640301; successor FH-WT1-E50 + azacitidine on NCT07645469)

Chapuis et al. *Nature Medicine* 2019

- **Disease context:High-risk AML in remission after allogeneic HCT** (maintenance) — 12 HLA-A2-positive patients transplanted for poor-risk AML and infused while in remission; contemporaneous comparison group of 88 similar-risk AML patients from the same programme.
- **n:** 12
- **Toxicities:CRS** (grade any): 0.0% (n=12); **Pulmonary toxicity** (grade any): 0.0% (n=12); **Acute GVHD** (grade 3): 8.3% (n=12); **Chronic GVHD** (grade any): 55.0% (n=12)
- **Efficacy:RFS** 100%; **RFS** 54%; **AE_rate** 0%

Mazziotta et al. *Nature Communications* 2025

- **Disease context:Relapsed and/or refractory AML after allogeneic HCT** (2L+) — 15 HLA-A2-positive patients with relapsed/refractory AML post-HCT enrolled 2013-2019. At infusion two had overt disease, five were MRD-positive, and the rest had no evaluable disease.
- **n:** 15
- **Toxicities:Dose-limiting toxicity** (grade >=3): 0.0% (n=15); **Product-related serious AE** (grade >=3): 0.0% (n=15); **Acute GVHD** (grade 3): 6.7% (n=15); **Chronic GVHD (moderate)** (grade any): 6.7% (n=15)
- **Efficacy:OS** 242 days; **OS_rate** 33%; **OS_rate** 20%; **RR** 26.7%

MDG1011 (PRAME-directed, HLA-A*02:01-restricted autologous TCR-T) and BPX-701 (PRAME TCR-T with an iCasp9 switch)

Thomas et al. *Cancers (Basel)* 2025

- **Disease context:Relapsed/refractory AML, MDS and multiple myeloma** (2L+) — 13 HLA-A*02:01-positive patients enrolled, 9 infused. Advanced disease stage and rapid progression in the R/R AML subset limited what could be read out.
- **n:** 9
- **Toxicities:Dose-limiting toxicity** (grade >=3): 0.0% (n=9); **CRS** (grade any): 22.2% (n=9)
- **Efficacy:RR** 22.2%; **biomarker** 77.8%

Cd123 interventions

1. Pivekimab sunirine (IMGN632, CD123-directed antibody-drug conjugate) at the 0.045 mg/kg RP2D, off-label after the May 2026 BPDCN approval

Likelihood of effect: Moderate as debulking: the only CD123 agent with own-disease data (ORR 21%, 95% CI 8-40; cCR 17%), but the interval carries little information on n=29.

Toxicity burden: Moderate (hepatic veno-occlusive disease at four to ten times the RP2D, both events reversible; at the RP2D grade 3+ febrile neutropenia 10%, infusion reactions 7%, anaemia 7%)

Counter-productive MoA:Moderate (Hepatic veno-occlusive disease before conditioning could cost the transplant the debulking is meant to enable.)

Overall:The only CD123 agent with data in her own disease and line, held under a narrowed hepatic condition rather than a bar: liver labs and weekly monitoring first.

Key references: PMID 38423051 · NCT04086264 · PMID 41671533

2. Tagraxofusp (SL-401, CD123-directed IL-3 / diphtheria-toxin fusion), off-label in AML or via the R/R AML cohort of NCT03113643

Likelihood of effect: Low-to-uncertain in AML: the headline 72% CR is first-line BPDCN cross-tumour, and the nearest AML data (69% ORR, n=26) are untreated disease.

Toxicity burden: High (capillary leak syndrome 19% with a fatal case in each dose subgroup, transaminitis ALT 64% / AST 60%, hypoalbuminaemia 55%)

Counter-productive MoA:Low (No mechanistic antagonism; on-target killing of normal CD34+CD123+ progenitors adds myelosuppression ahead of a graft.)

Overall:The only labelled CD123 agent with a real capillary-leak management algorithm, but its headline number is BPDCN cross-tumour and it needs an albumin and cardiac baseline first.

Key references: PMID 31018069 · PMID 38052038 · NCT03113643

Also considered — not ranked

Feature-targeting investigational options the board considered but did not rank as live top-tier choices: Unavailable (program discontinued), Consolidated (one product of an approach ranked above), Thin evidence (no peer-reviewed clinical efficacy yet), or Not enrollable (cross-tumor or geographically inaccessible).

CD123-directed CAR-T and switchable CD123 CAR platforms (Allo-RevCAR01-T + R-TM123, NCT05949125) — _Thin evidence_

Likelihood of effect: Unknown: no published clinical efficacy for the construct. Recruiting, with a switchable design that addresses on-target myelosuppression.

Toxicity burden: Moderate (no published grade profile; CAR-T class cytokine release syndrome and lymphodepletion cytopenias expected)

Counter-productive MoA:Low (On-target killing of normal CD123+ progenitors adds myelosuppression; the switchable module is designed to limit it.)

Overall:A switchable off-the-shelf CD123 CAR that is recruiting and eligible on disease state, with no published clinical efficacy to rank on.

Key references: NCT05949125

VIP943 (CD123-directed antibody-drug conjugate, NCT06034275) — _Unavailable_

Likelihood of effect: Not reliably obtainable: sponsor wind-down in 2025 with a stale registry entry, despite eligibility criteria that fit her unusually well.

Toxicity burden: Low (no published clinical toxicity)

Counter-productive MoA:Low (CD123-ADC on-target myelosuppression; no mechanism objection raised.)

Overall:A CD123 ADC whose posted criteria fit her better than most, on a programme the sponsor began winding down in 2025.

Key references: NCT06034275

Flotetuzumab (MGD006, CD123 x CD3 DART) — _Unavailable_

Likelihood of effect: Not obtainable (discontinued): CR/CRh 26.7% in exactly her chemorefractory population; the 47% TP53 figure is post-hoc on 15 patients.

Toxicity burden: Moderate (infusion-related reactions or cytokine release in 96%, grade 3+ in 8%, managed with step-up dosing and tocilizumab)

Counter-productive MoA:Moderate (CD123xCD3 redirection needs fit T cells; post-transplant and post-FLAG-IDA exhaustion may blunt it. TP53 rationale is post-hoc.)

Overall:A discontinued drug carrying the most relevant salvage number in the file, and the reason to chase an open CD123 or peptide-HLA engager instead.

Key references: PMID 32929488 · PMID 33057635 · NCT02152956

Vibecotamab (XmAb14045, CD123 x CD3 bispecific) — _Unavailable_

Likelihood of effect: Not obtainable (terminated): ORR 9.0% on n=120, the weakest CD123 redirection signal in the file, with 26.7% grade 3+ cytokine release.

Toxicity burden: High (cytokine release syndrome 85% any grade, 26.7% grade 3+ including one grade 5)

Counter-productive MoA:Moderate (Same redirection mechanism as flotetuzumab with a weaker signal and a heavier cytokine-release burden.)

Overall:The second CD123 bispecific to stop, with the weakest activity and the heaviest cytokine-release burden of the class.

Key references: PMID 37647601 · NCT05285813

Evidence in detail

Pivekimab sunirine (IMGN632, CD123-directed antibody-drug conjugate) at the 0.045 mg/kg RP2D, off-label after the May 2026 BPDCN approval

Daver et al. *Lancet Oncology* 2024

- **Disease context:Relapsed or refractory CD123-positive AML** (2L+) — 91 enrolled overall (68 on the every-3-week schedule A); the efficacy estimate below is the 29 patients treated at the recommended phase 2 dose. ECOG 0-1 required.
- **n:** 29
- **Toxicities:Febrile neutropenia** (grade >=3): 10.0% (n=29); **Infusion-related reaction** (grade >=3): 7.0% (n=29); **Anaemia** (grade >=3): 7.0% (n=29); **Veno-occlusive disease** (grade >=3): 2.9% (n=68); **Treatment-related death** (grade 5): 1.5% (n=68)
- **Efficacy:ORR** 21% (95% CI 8-40); **CR_rate** 17% (95% CI 6-36)

Pemmaraju et al. *Journal of Clinical Oncology* 2026

- **Disease context:Blastic plasmacytoid dendritic cell neoplasm, frontline and relapsed/refractory** (any) — 84 patients: 33 frontline (22 de novo, 20 forming the primary analysis population) and 51 relapsed/refractory. Median age 72.
- **n:** 84
- **Toxicities:Peripheral oedema** (grade any): 54.0% (n=84); **Peripheral oedema** (grade >=3): 12.0% (n=84); **Fatigue** (grade any): 26.0% (n=84); **Infusion-related reaction** (grade any): 26.0% (n=84); **Neutropenia** (grade >=3): 16.0% (n=84); **Thrombocytopenia** (grade >=3): 14.0% (n=84); **Pneumonia** (grade >=3): 6.0% (n=84)
- **Efficacy:CR_rate** 75% (95% CI 51-91); **CR_rate** 14% (95% CI 6-26); **DoR** 10.6 months; **DoR** 9.2 months; **OS** 16.6 months; **OS** 5.8 months (95% CI 3.9-8.4)

Tagraxofusp (SL-401, CD123-directed IL-3 / diphtheria-toxin fusion), off-label in AML or

Pemmaraju et al. *New England Journal of Medicine* 2019

- **Disease context:Blastic plasmacytoid dendritic cell neoplasm, untreated or relapsed** (1L) — 47 patients assigned across 7 and 12 microg/kg cohorts; the primary-outcome population was the 29 previously untreated patients dosed at 12 microg/kg. Median age 70.
- **n:** 29
- **Toxicities:Capillary leak syndrome** (grade any): 19.0% (n=47); **ALT increased** (grade any): 64.0% (n=47);

- **AST increased** (grade any): 60.0% (n=47); **Hypoalbuminaemia** (grade any): 55.0% (n=47); **Peripheral oedema** (grade any): 51.0% (n=47); **Thrombocytopenia** (grade any): 49.0% (n=47)
- **Efficacy:CR_rate** 72%; **ORR** 90%; **ORR** 67%; **OS** 8.5 months; **OS_rate** 59%; **OS_rate** 52%

Lane et al. *Blood Advances* 2024

- **Disease context:CD123-positive AML and high-risk MDS** (1L) — 56 adults enrolled; the untreated-AML triplet expansion was 26 patients, median age 71, all ELN adverse-risk, half TP53-mutated. 13 of the 26 carried a TP53 mutation.
- **n:** 26
- **Toxicities:Capillary leak syndrome** (grade any): 18.9% (n=37); **Capillary leak syndrome** (grade any): 47.0% (n=19); **Thrombocytopenia** (grade >=3): 45.9% (n=37); **White blood cell count decreased** (grade >=3): 45.9% (n=37); **Neutrophil count decreased** (grade >=3): 37.8% (n=37); **Anaemia** (grade >=3): 27.0% (n=37); **Febrile neutropenia** (grade >=3): 24.3% (n=37); **Infection** (grade >=3): 16.2% (n=37); **Tumour lysis syndrome** (grade >=3): 10.8% (n=37); **30-day all-cause mortality** (grade 5): 10.8% (n=37)
- **Efficacy:ORR** 69%; **CR_rate** 39%; **ORR** 54%; **OS** 14 months; **PFS** 8.5 months; **biomarker** 71%

Flotetuzumab (MGD006, CD123 x CD3 DART)

Uy et al. *Blood* 2021

- **Disease context:Relapsed/refractory AML, primary induction failure or early relapse** (2L+) — 88 adults with R/R AML overall (42 dose-finding, 46 at the recommended phase 2 dose). The efficacy estimate is the 30 primary-induction-failure or early-relapse patients treated at the recommended dose.
- **n:** 30
- **Toxicities:IRR/CRS** (grade any): 96.0% (n=50); **IRR/CRS** (grade >=3): 8.0% (n=50); **Anaemia** (grade >=3): 28.4% (n=88); **Thrombocytopenia** (grade >=3): 20.5% (n=88); **Lymphopenia** (grade >=3): 18.2% (n=88); **Hypophosphataemia** (grade >=3): 14.8% (n=88); **Hypokalaemia** (grade >=3): 13.6% (n=88); **Treatment-related death** (grade 5): 0.0% (n=88)
- **Efficacy:CR_rate** 26.7%; **ORR** 30.0%; **OS** 10.2 months; **OS_rate** 75% (95% CI 45.0-100.0); **OS_rate** 50% (95% CI 15.4-84.6)

Vadakekolathu et al. *Blood Advances* 2020

- **Disease context:Relapsed/refractory AML with TP53 mutation and/or 17p deletion** (2L+) — 15 of 45 flotetuzumab-treated patients with marrow transcriptomic profiling carried TP53 mutations and/or 17p deletion. Responders had higher baseline tumour-inflammation, FOXP3, CD8 and PD1 signature scores.
- **n:** 15
- **Toxicities:** —
- **Efficacy:CR_rate** 47%; **OS** 10.3 months

Vibecotamab (XmAb14045, CD123 x CD3 bispecific)

Ravandi et al. *Blood Advances* 2023

- **Disease context:Relapsed/refractory AML** (2L+) — 120 patients dosed, 111 AML patients evaluable for response; 87 evaluable at the optimised dose of at least 0.75 microg/kg.
- **n:** 120
- **Toxicities:CRS** (grade any): 85.0% (n=120); **CRS** (grade >=3): 26.7% (n=120); **Any treatment-emergent AE** (grade >=3): 87.5% (n=120); **Anaemia** (grade >=3): 25.8% (n=120); **Febrile neutropenia** (grade >=3): 25.0%

(n=120); **Pneumonia** (grade >=3): 17.5% (n=120); **Treatment-related death** (grade 5): 0.8% (n=120)
- **Efficacy:ORR** 9.0%; **ORR** 11.5%; **CR_rate** 5.4%

Tp53 Y220C interventions

1. Rezatapopt (PC14586, TP53 Y220C-selective p53 reactivator) + azacitidine (NCT06616636) - considered and declined

Likelihood of effect: Not assessable: no peer-reviewed clinical effect size in myeloid disease, and the only mechanistic study needed venetoclax for killing - the drug she just failed.

Toxicity burden: Low (none extractable - no peer-reviewed clinical toxicity in myeloid disease; azacitidine-backbone cytopenias apply)

Counter-productive MoA:Moderate (Reactivated Y220C p53 restored conformation with limited or no apoptosis unless venetoclax was added, the agent she just progressed through.)

Overall:Declined, not dropped: double-gated on a Y220C call at the detection limit of the panel that called her negative, and needing venetoclax to kill.

Key references: NCT06616636 · PMID 40608889

Also considered — not ranked

Feature-targeting investigational options the board considered but did not rank as live top-tier choices: Unavailable (program discontinued), Consolidated (one product of an approach ranked above), Thin evidence (no peer-reviewed clinical efficacy yet), or Not enrollable (cross-tumor or geographically inaccessible).

Eprenetapopt (APR-246) plus venetoclax and azacitidine, and as post-transplant maintenance — _Unavailable_

Likelihood of effect: Not obtainable (discontinued after a negative phase 3): single-arm CR 38% in treatment-naive TP53-mutant AML did not replicate randomised.

Toxicity burden: High (grade 3+ febrile neutropenia 47%, thrombocytopenia 37%, one treatment-related death)

Counter-productive MoA:Low (No goal-specific antagonism; the failure mode is efficacy that did not replicate, not mechanism working against the goal.)

Overall:The pan-p53 reactivator whose single-arm signal did not survive randomisation - explicitly not the fallback if the Y220C call comes back negative.

Key references: PMID 36990622 · PMID 35816664 · NCT04214860

Evidence in detail

Rezatapopt (PC14586, TP53 Y220C-selective p53 reactivator) + azacitidine (NCT06616636) - considered and declined

Carter et al. *Blood* 2025

- **Disease context:** —
- **n:** —
- **Toxicities:** —
- **Efficacy:** —

Eprenetapopt (APR-246) plus venetoclax and azacitidine, and as post-transplant maintenance

Garcia-Manero et al. *Lancet Haematology* 2023

- **Disease context:TP53-mutated AML, treatment-naive** (1L) — 49 enrolled across cohorts; the response estimate is the 39 patients on the triplet. At least one pathogenic TP53 mutation was required, ECOG 0-2, median age 67.
- **n:** 39
- **Toxicities:Febrile neutropenia** (grade >=3): 47.0% (n=49); **Thrombocytopenia** (grade >=3): 37.0% (n=49); **Leukopenia** (grade >=3): 25.0% (n=49); **Anaemia** (grade >=3): 22.0% (n=49); **Treatment-related serious AE** (grade >=3): 27.0% (n=49); **Treatment-related death (sepsis)** (grade 5): 2.0% (n=49)
- **Efficacy:CR_rate** 38% (95% CI 23-55); **ORR** 64% (95% CI 47-79)

Mishra et al. *Journal of Clinical Oncology* 2022

- **Disease context:TP53-mutated AML and MDS after allogeneic HCT** (maintenance) — 33 of 55 transplanted patients started maintenance (14 AML, 19 MDS), median age 65. Screening began before transplant, so the cohort is enriched for patients who reached and tolerated HCT.
- **n:** 33
- **Toxicities:Acute GVHD** (grade any): 12.0% (n=33); **Chronic GVHD** (grade any): 33.0% (n=33); **60-day mortality** (grade 5): 6.0% (n=33); **30-day mortality** (grade 5): 0.0% (n=33)
- **Efficacy:RFS** 12.5 months; **RFS** 59.9% (95% CI 41.0-74.0); **OS** 20.6 months; **OS_rate** 78.8% (95% CI 60.6-89.3)

Wt1 interventions

Also considered — not ranked

Feature-targeting investigational options the board considered but did not rank as live top-tier choices: Unavailable (program discontinued), Consolidated (one product of an approach ranked above), Thin evidence (no peer-reviewed clinical efficacy yet), or Not enrollable (cross-tumor or geographically inaccessible).

WT1-siTCR gene-transduced lymphocytes (TBI-1301), HLA-A*24:02-restricted — _Not enrollable_

Likelihood of effect: Not enrollable: Japan-only development with no US or EU route. Route-preservation only - transient blast reductions in 2 of 8.

Toxicity burden: Low (no normal-tissue adverse events reported across n=8)

Counter-productive MoA:Low (The siTCR design silences the endogenous TCR and spared normal tissue in the published series.)

Overall:The A*24:02 fallback that keeps an immunotherapy thread alive if A*02:01 types negative, with no route a US patient can currently enter.

Key references: PMID 28860210

Donor-derived and off-the-shelf multi-leukaemia-antigen-specific T cells (TAA-T on NCT02203903, ADSPAM on NCT02494167, MT-401-OTS on NCT06552416) — _Not enrollable_

Likelihood of effect: Not enrollable: the open protocols exclude post-transplant relapse, a planned second HCT, or cap blasts at 10%. Modest activity (2/8) in active disease.

Toxicity burden: Low (no grade >2 acute GVHD and no extensive chronic GVHD across n=25)

Counter-productive MoA:Low (Multi-antigen targeting limits single-antigen escape; no mechanism that blunts the goal.)

Overall:The HLA-independent hedge that hits WT1 and PRAME at once, locked behind disease-state, age and second-transplant exclusions on every open protocol.

Key references: PMID 33270816 · NCT02494167 · NCT06552416

Evidence in detail

WT1-siTCR gene-transduced lymphocytes (TBI-1301), HLA-A*24:02-restricted

Tawara et al. *Blood* 2017

- **Disease context:Refractory AML and high-risk MDS** (2L+) — 8 HLA-A*24:02-positive patients with refractory AML or high-risk MDS across two dose cohorts. Restriction is A*24:02, so this route stays open if A*02:01 typing returns negative.
- **n:** 8
- **Toxicities:Normal-tissue adverse event** (grade any): 0.0% (n=8)
- **Efficacy:RR** 25%; **biomarker** 62.5%; **OS_rate** 80%

Donor-derived and off-the-shelf multi-leukaemia-antigen-specific T cells (TAA-T on NCT02203903, ADSPAM on NCT02494167, MT-401-OTS on NCT06552416)

Lulla et al. *Blood* 2021

- **Disease context:AML/MDS after allogeneic HCT, high relapse risk or overt relapse** (2L+) — 25 patients infused, 17 at high risk of relapse and 8 with relapsed disease. Products were generated from 29 HCT donors and were not HLA-restricted to a single allele.
- **n:** 25
- **Toxicities:Acute GVHD** (grade >=3): 0.0% (n=25); **Chronic GVHD (extensive)** (grade any): 0.0% (n=25)
- **Efficacy:ORR** 25%; **PFS** not reached months; **OS** not reached months

Ha1 interventions

Also considered — not ranked

Feature-targeting investigational options the board considered but did not rank as live top-tier choices: Unavailable (program discontinued), Consolidated (one product of an approach ranked above), Thin evidence (no peer-reviewed clinical efficacy yet), or Not enrollable (cross-tumor or geographically inaccessible).

GLIDE (guided lymphocyte immunopeptide-derived expansion; donor anti-minor-histocompatibility-antigen T cells, NCT03091933) — _Unavailable_

Likelihood of effect: Not obtainable: registry status unknown, academic programme with no filing. Design is the only HLA-agnostic minor-antigen route in the dossier.

Toxicity burden: Low (no published clinical toxicity)

Counter-productive MoA:Low (Minor-antigen T cells carry GVHD risk in principle; no data either way.)

Overall:The one minor-antigen approach that does not bottleneck on A*02:01, with no current route and an unknown registry status.

Key references: NCT03091933